

LADR 4e — Alternative Solutions

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This document collects alternative proofs for selected exercises in *Linear Algebra Done Right* (4th edition) by Sheldon Axler. Each solution here differs from the official or standard approach and aims to be more direct or illuminating.

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Chapter 6.B

Exercise (6.B.9). Suppose $e_1, \dots, e_m$ is the result of applying the Gram–Schmidt procedure to a linearly independent list $v_1, \dots, v_m \in V$. Prove that $\langle v_k, e_k \rangle > 0$ for each $k = 1, \dots, m$.

Solution. By the Gram–Schmidt procedure, $e_k = \frac{f_k}{\|f_k\|}$, so

$$\langle v_k, e_k \rangle = \frac{\langle v_k, f_k \rangle}{\|f_k\|}.$$

It suffices to show that $\langle v_k, f_k \rangle > 0$.

Substituting the definition

$$f_k = v_k - \sum_{i=1}^{k-1} \frac{\langle v_k, f_i \rangle}{\|f_i\|^2} f_i$$

and using $f_i = \|f_i\| e_i$, we obtain

$$\langle v_k, f_k \rangle = \|v_k\|^2 - \sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2.$$

By Bessel’s inequality,

$$\sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2 \leq \|v_k\|^2,$$

so $\langle v_k, f_k \rangle \geq 0$.

If equality holds, then $\|v_k\|^2 = \sum_{i=1}^{k-1} |\langle v_k, e_i \rangle|^2$. By Exercise 6.B.3, this implies $v_k \in \text{span}(e_1, \dots, e_{k-1}) = \text{span}(v_1, \dots, v_{k-1})$, contradicting the linear independence of $v_1, \dots, v_m$.

Therefore $\langle v_k, f_k \rangle > 0$, and hence

$$\langle v_k, e_k \rangle = \frac{\langle v_k, f_k \rangle}{\|f_k\|} > 0.$$

Chapter 7.F

Exercise (7.F.3). Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Prove that

$$\|Tv\| = \|T\| \|v\| \iff T^*Tv = \|T\|^2v.$$

Solution. ($\Leftarrow$) Suppose $T^*Tv = \|T\|^2v$. Taking the inner product of both sides with v gives

$$\langle T^*Tv, v \rangle = \langle \|T\|^2v, v \rangle.$$

The left-hand side equals $\langle Tv, Tv \rangle = \|Tv\|^2$, and the right-hand side equals $\|T\|^2\|v\|^2$. Hence $\|Tv\|^2 = \|T\|^2\|v\|^2$, and taking square roots yields $\|Tv\| = \|T\| \|v\|$.

($\Rightarrow$) Suppose $\|Tv\| = \|T\| \|v\|$. Let $S = \|T\|^2I - T^*T$. We first show that S is a positive operator. For each $u \in V$,

$$\langle Su, u \rangle = \|T\|^2\|u\|^2 - \langle T^*Tu, u \rangle = \|T\|^2\|u\|^2 - \|Tu\|^2 \geq 0,$$

where the inequality follows from $\|Tu\| \leq \|T\| \|u\|$. Since T^*T and I are self-adjoint, so is S ; thus S is positive. Next, using the hypothesis $\|Tv\| = \|T\| \|v\|$,

$$\langle Sv, v \rangle = \|T\|^2\|v\|^2 - \|Tv\|^2 = 0.$$

By 7.43, a positive operator S with $\langle Sv, v \rangle = 0$ satisfies $Sv = 0$. Therefore $\|T\|^2v - T^*Tv = 0$, that is,

$$T^*Tv = \|T\|^2v.$$

Exercise (7.F.8). (a) Prove that if $T \in \mathcal{L}(V)$ and $\|I - T\| < 1$, then T is invertible.

(b) Suppose that $S \in \mathcal{L}(V)$ is invertible. Prove that if $T \in \mathcal{L}(V)$ and

$$\|S - T\| < \frac{1}{\|S^{-1}\|},$$

then T is invertible.

Solution. We prove (b) directly; part (a) follows as the special case $S = I$.

Since V is finite-dimensional, it suffices to show that T is injective, i.e., $Tv \neq 0$ whenever $v \neq 0$.

We first observe that S admits a uniform lower bound on its stretching: for every $v \in V$,

$$\|v\| = \|S^{-1}Sv\| \leq \|S^{-1}\| \|Sv\|,$$

which rearranges to

$$\|Sv\| \geq \frac{1}{\|S^{-1}\|} \|v\|. \quad (*)$$

Now let $v \in V$ with $v \neq 0$. Writing $T = S - (S - T)$ gives

$$Tv = Sv - (S - T)v.$$

By the reverse triangle inequality,

$$\|Tv\| \geq \|Sv\| - \|(S - T)v\|.$$

Applying (*) to the first term and $\|(S - T)v\| \leq \|S - T\| \|v\|$ to the second,

$$\|Tv\| \geq \frac{1}{\|S^{-1}\|} \|v\| - \|S - T\| \|v\| = \left(\frac{1}{\|S^{-1}\|} - \|S - T\| \right) \|v\|.$$

By hypothesis $\|S - T\| < \frac{1}{\|S^{-1}\|}$, so the coefficient on the right is strictly positive. Since $\|v\| > 0$, we conclude $\|Tv\| > 0$, hence $Tv \neq 0$.

Therefore T is injective, and thus invertible.